



JIT2321 - DIGITAL PRINCIPLES AND COMPUTER ORGANISATION (DPCO)

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Computer Architecture VS Computer Organization

Computer Architecture	Computer Organization
Concerned with the way hardware components are connected together to form a computer system.	Concerned with the structure and behaviour of a computer system as seen by the user.
Acts as the interface between hardware and software.	Deals with the components of a connection in a system.
Helps us to understand the functionalities of a system.	Tells us how exactly all the units in the system are arranged and interconnected.
A programmer can view architecture in terms of instructions, addressing modes and registers.	Whereas Organization expresses the realization of architecture.
While designing a computer system architecture is considered first.	An organization is done on the basis of architecture.
Deals with high-level design issues.	Deals with low-level design issues.
Architecture involves Logic (Instruction sets, Addressing modes, Data types, Cache optimization)	Organization involves Physical Components (Circuit design, Adders, Signals, Peripherals)

NUMBER SYSTEM

- It is defined as a system of writing to express numbers.
- It is the mathematical notation for representing numbers of a given set by using digits or other symbols in a consistent manner.
- The value of any digit in a number can be determined by:
 - The digit
 - Its position in the number
 - The base of the number system

Characteristics of Numbering Systems

- 1) The digits are **consecutive**.
- 2) The number of digits is equal to the size of the base.
- 3) Zero is always the first digit.
- 4) The base number is never a digit.
- 5) When 1 is added to the largest digit, a sum of zero and a carry of one results.
- 6) Numeric values are determined by the implicit **positional values** of the digits.

Base-N Number System

- Base N
- N Digits: 0, 1, 2, 3, 4, 5, ..., N-1
- Example: 1045_N
 - Positional Number System

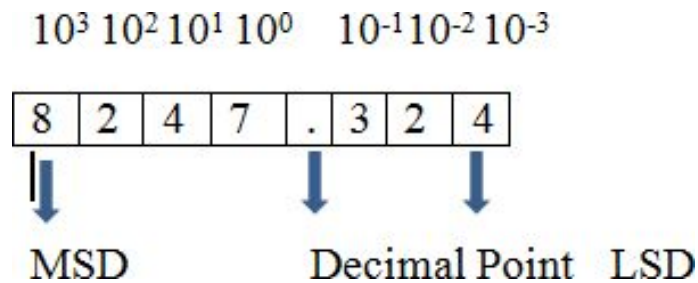
$$\begin{array}{ccccccc} N^{n-1} & \cdots & N^4 & N^3 & N^2 & N^1 & N^0 \\ & & & & & & d_0 \\ & & & & & & \vdots \\ & & & & & & d_1 \end{array}$$

- 
- Digit d_0 is the least significant digit (LSD).
 - Digit d_{n-1} is the most significant digit (MSD).
 - The most common number systems used in digital technology. They are,
 - Decimal number system
 - Binary number system
 - Octal number system
 - Hexadecimal number system

Decimal Number System

- Composed of ten symbols (0, 1, 2, 3, 4, 5, 6, 7, 8, 9) known as Base 10 system. Example: 1045_{10}
- Positional Value System in which value of digit depends on its position.
- **Example:** 8247.324 is represented as

$$(8 \times 10^3) + (2 \times 10^2) + (4 \times 10^1) + (7 \times 10^0) + (3 \times 10^{-1}) + (2 \times 10^{-2}) + (4 \times 10^{-3})$$



Binary Number System

- Composed of two symbols (0,1) also called as “**Base 2 system**”
- Used to model the series of electrical signals computers use to represent information
- 0 represents no voltage or an **off state** and 1 represents the presence of voltage or an **on state**. Binary numbers 0 and 1 known as bits.
- **Example:** 1011.101 is represented as

$$(1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3})$$

$2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3}$

1	0	1	1	.	1	0	1
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MSD



Decimal Point



LSD

Octal Number System

- Composed of eight symbols (0,1,2,3,4,5,6,7) also known as the **Base 8 System**
- Groups of three (binary) digits can be used to represent each octal digit
- Compared with binary system this is compact and occupies less space for data.
- **Example:** 5432 is represented as
$$5 \times 8^3 + (4 \times 8^2) + (3 \times 8^1) + (2 \times 8^0)$$

Hexa Decimal Number System

- Base 16 system
- Uses 16 symbols (digits 0-9 & letters A,B,C,D,E,F)
- Groups of four bits represent each base 16 digit

Example: 306B.1C is represented as

$$(3 \times 16^3) + (0 \times 16^2) + (6 \times 16^1) + (B \times 16^0) + (1 \times 16^{-1}) + (C \times 16^{-2})$$

FOUR BASIC NUMBER SYSTEM DIGITS TABLE

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

CONVERSIONS

- Conversion of a number system are, decimal to any base , any base to decimal, binary to any base, any base to binary, octal to hexa, hexa to octal.

Decimal to any base conversion

a) Decimal to binary conversion

- i) Converting a decimal number by 2 until quotient of 0 is obtained.
- ii) Binary number is obtained by taking reminder after each division in reverse order. (double-dabble method)



Example:

89.625

Integer Part

Fraction Part

$$(89.625)_{10} = (\quad)_2$$

2	89		
2	44	1	↑
2	22	0	↑
2	11	0	↑
2	5	1	↑
2	2	1	↑
2	1	0	↑
	0	1	↑

$$0.625 \times 2 = 1.250 \Rightarrow 1 \downarrow$$

$$0.250 \times 2 = 0.500 \Rightarrow 0 \downarrow$$

$$0.500 \times 2 = 1.0 \Rightarrow 1 \downarrow$$

$$\therefore (89.625)_{10} = (1011001.101)_2$$

- Binary number is obtained by multiplying number continuously by 2, recording a carry each time is known as fraction conversion.

b) Decimal to Octal conversion

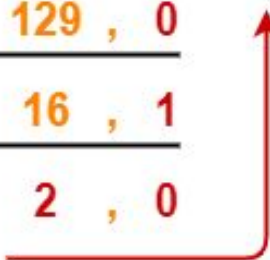
- Procedure is same as that of double dabble method

Example:

$$(1032.6875)_{10}$$

$$(1032.6875)_{10} \rightarrow (?)_8$$

8	1032
8	129 , 0
8	16 , 1
	2 , 0



For Real Part-

- We convert the real part from base 10 to base 8 using division method same as above.
- So, $(1032)_{10} = (2010)_8$

For Fractional Part-

- We convert the fractional part from base 10 to base 8 using multiplication method.

Step-01:

- Multiply 0.6875 with 8. Result = 5.5.
- Write 5 in real part and 0.5 in fractional part.

Step-02:

- Multiply 0.5 with 8. Result = 4.0.
- Write 4 in real part and 0.0 in fractional part.
- Since fractional part becomes 0, so we stop.

	Real part	Fractional Part
0.6875×8	5	0.5
0.5×8	4	0.0

- From here, $(0.6875)_{10} = (0.54)_8$
- Combining the result of real and fractional parts, we have,

$$(1032.6875)_{10} = (2010.54)_8$$

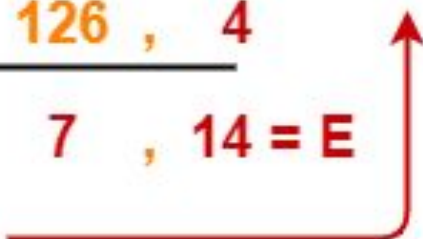
c) Decimal to Hexa conversion

Same as double-dabble method.

Example: $(2020)_{10}$

● Using division method, we have-

16	2020
16	126 , 4
	7 , 14 = E



Any base to Decimal conversion

a) Binary to Decimal

Example: $(10111.1101)_2 \rightarrow (?)_{10}$

- Converted into decimal by multiplying 1 or 0 by their weight adding the two integer conversion.

Integer Part : 10111

$$= (1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0)$$

$$= (32 + 0 + 8 + 4 + 2 + 1)$$

$$= (47)$$

Fraction Part

1101

$$= (1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4})$$

$$= (0.5000 + 0.2500 + 0.0000 + 0.0625)$$

$$= 0.8125$$

$$\text{Result} = (101111.1101)_2 \rightarrow (47.8125)_{10}$$

b) Octal to
Decimal Example:
 $(36.4)_8$

Integer Conversion : 36
 $= 3 \times 8^1 + 6 \times 8^0$
 $= 24 + 6$
 $= 30$

Fraction Conversion : 0.4
 $= 4 \times 8^{-1}$
 $= 4 \times 0.125$
 $= 0.500$

Result : $(36.4)_8 = (30.5)_{10}$

c) Hexa to Decimal

Example:

$$(A3B)_{16} \\ = A \times 16^2 + 3 \times 16^1 + B \times 16^0$$

$$= 2560 + 048 + 011$$

$$= 2619$$

Result :

$$(A3B)_{16} = (2619)_{10}$$

Binary to any Base Conversion

a) Binary to Octal

Reverse of Octal to Binary

Procedure: Form 3-bit combination moving from right most position to left position

Example: $(10011010101)_2$

010 011 010 101

2 3 2 5

$$(10011010101)_2 = (2325)_8$$

b) Binary to Hexa

Procedure: Write binary number starting from right most position, the group binary number into groups of four bits.

Example:

$$(10100110)_2$$

$$1010 \quad 0110$$

$$A \quad 6$$

$$(10100110)_2 = (A6)_{16}$$

Any Base to Binary Conversion

a) Octal to Binary

Reverse method of binary to octal.

Example:

$$(532)_8$$

5	3	2
101	011	010

$$(532)_8 = (101011010)_2$$

b) Hexa to Binary

Reverse method of binary to hexa conversion.

Example:

	2	D	6
$(2D6)_{16}$			
	0010	1101	0110

$$(2D6) = (001011010110)_2$$

Hexadecimal & Conversion to Octal

Convert the given hexadecimal number to binary equivalent and then from binary to octal.

Example:

$$(5C2)_{16}$$

(Hexa to Binary)	5	C	2	
	0101	1100	0010	
(Binary to Octal)	010	111	000	010
	2	7	0	2

$$(5C2) = (2702)_8$$

b) Octal to Hexa

- Convert octal to binary number and binary into equivalent hexadecimal number.

Example:

$$(321)_8$$

Octal to Binary

3

2

1

011

010

001

Binary to Hexa

1101

0001

D

1

$$(321)_8 = (D1)_{16}$$

BOOLEAN ALGEBRA

- The boolean algebra is used to simplify the design the logic circuits and involves lengthy mathematical operations.
- Binary logic deals with two values 0 and 1.(0-False, 1-True)
- Karnaugh Map- Simplification of boolean equation upto four equations. (Difficult if more than five input variables).
- Quine Mcclusky method- Tabular method of minimization(reduces the requirement of hardware)

Boolean Logic Operations

- Boolean Function- Algebraic expression formed using constant, binary variables and basic logic operation symbols.
-
- Logical Operation
 1. AND (Multiplication)
 2. OR (Addition)
 3. NOT (Logical Complementation)

Logical AND

$$Y=A.B$$

Truth
Table

Input		Output
A	B	$Y=A.B$
0	0	0
0	1	0
1	0	0
1	1	1

Logical OR

$$Y=A+B$$

Truth Table

Input		Output
A	B	$Y=A+B$
0	0	0
0	1	1
1	0	1
1	1	1

Logical NOT (Inverter)

Convert logical 1 to 0 and vice versa.

Truth Table

Input	Output
A	Y
0	1
1	0

Basic Laws of Boolean Algebra

Logical operations expressed and mathematically using the rules, laws minimized and boolean algebra. theorems of

Boolean Addition:

$$A+0=A$$

$$A+1=1$$

$$A+A=A$$

$$A + \bar{A}$$

$$=1 \quad \bar{A}=$$

$$\bar{A}$$

$$1 + \bar{A}=1$$



Boolean Multiplication:

$$A.1=A$$

$$A.0=0$$

$$A.A=A$$

$$A.\bar{A}=0$$

$$((A)')'=A$$

Properties of Boolean Algebra

- Commutative Properties
- Associative Properties
- Idempotent Properties
- Identity Properties
- Null Properties
- Distributive Properties
- Negation Properties
- Double Negation Properties
- Absorption Properties
- Demorgan's Theorem

1) Commutative Properties

● Boolean Addition

$$A+B=B+A$$

Proof

A	B	A+B	B+A
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

- **Boolean Multiplication**

- **$A.B=B.A$**

Proof

A	B	A.B	B.A
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

2) Associative Properties

$$\mathbf{A}+(\mathbf{B}+\mathbf{C})=(\mathbf{A}+\mathbf{B})+\mathbf{C}$$

Proof

A	B	C	A+B	B+C	(A+B)+C	A+(B+C)
0	0	0	0	0	0	0
0	0	1	0	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	0	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

- 
- Similarly, the associative law of multiplication is given by,

$$A.(B.C)=(A.B).C$$

3) Distributive Properties

i) Boolean Addition

It is distributive over boolean multiplication.

$$A+BC=(A+B).(A+C)$$

● Proof

LHS:

$$A+BC = A.1+BC$$

[Since $A.1=A$]

$$=A.(1+B)+BC$$

[Since $1+B=1$]

$$=A.1+AB+BC$$

$$=A(1+C)+AB+BC$$

[Since $1+C=1$]

$$=A.1+AC+AB+BC$$

$$=A+AC+AB+BC$$

$$=A.A+AC+AB+BC$$

[Since $A.A=A$]

$$=A(A+C)+B(A+C)$$

$$=(A+C)(A+B)$$

$$=RHS$$

ii) Boolean Multiplication

Distributive over boolean

addition $A.(B+C)=A.B+A.C$

Proo

f

$$\begin{aligned} \text{LHS} &= A.B + A.C \\ &= A.B(1+C) + A.C(1+B) && [\text{Since } 1+C=1, 1+B=1] \\ &= A.B + ABC + A.C + A.B.C \\ &= A.B + A.C + ABC && [\text{Since } A+A=A] \\ &= AB(1+C) + AC && [\text{Since } 1+C=1] \\ &= AB + AC \\ &= A.(B+C) \\ &= \text{RHS} \end{aligned}$$

4) Idempotent Properties

i) $A + A = A$

Proof

If $A=0$ then $0+0=0=A$

If $A=1$ then $1+1=1=A$

ii) $A.A = A$

Proof

If $A=0$ then $0.0=0=A$

If $A=1$ then $1.1=1=A$

5) Identity Properties

i) $A \cdot 1 = A$

Proof

If $A=0$ then $0 \cdot 1 = 0 = A$

If $A=1$ then $1 \cdot 1 = 1 = A$

ii) $A + 1 = 1$

Proof

If $A=0$ then $0 + 1 = 1$

If $A=1$ then $1 + 1 = 1$

6) Null Properties

i) $A \cdot 0 = 0$

Proof

If $A=0$ then $0 \cdot 0 = 0$

If $A=1$ then $1 \cdot 0 = 0$

ii) $A + 0 = A$

Proof

If $A=0$ then $0 + 0 = 0 = A$

If $A=1$ then $0 + 1 = 1 = A$

7) Negative Properties

i) $A.A' = 0$

Proof

If $A=0$ then $0.1=0$

If $A=1$ then $1.0=0$

ii) $A + A' = 1$

Proof

If $A=0$ then $0+1=1$

If $A=1$ then $1+0=1$

8) Double Negation Properties

i) $((A)')' = A$

Proof

If $A=0$ then $(A)' = 1$, $((A)')' = 0 = A$

If $A=1$ then $(A)' = 0$, $((A)')' = 1 = A$

9) Absorption Properties

i) $A + AB = A$

Proof

$$\begin{aligned} \text{LHS: } A(1+B) &= A.1 && [\text{Since } 1+B=1] \\ &= A && [\text{Since } 1.A=A] \\ &= \text{RHS} \end{aligned}$$

ii) $A(A+B)=A$

Proof

$$\begin{aligned} \text{LHS: } A(A+B) &= A(A+B) \\ &= A.A+AB && [\text{Since } A.A=A] \\ &= A+AB \\ &= A(1+B) && [\text{Since } 1+B=1] \\ &= A \\ &= \text{RHS} \end{aligned}$$



iii)

$$A + A'B = A + B$$

Proof

$$\begin{aligned} \text{LHS: } A + A'B &= (A + A')(A + B) && [\text{Since } A + BC = (A + B)(A + C)] \\ &= 1(A + B) && [\text{Since } A + A' = 1] \\ &= A + B \\ &= \text{RHS} \end{aligned}$$

10) Demorgan's Theorem

- First Theorem

States that complement of a product is equal to be sum of the complements.

$$(A.B)' = A' + B'$$

- Second Theorem

States that complement of a sum equal to product of the complements. $(A+B)' = A'.B'$

● DeMorgan's First Theorem using Truth Table

Inputs		Truth Table Outputs For Each Term				
B	A	$A.B$	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A} + \overline{B}$
0	0	0	1	1	1	1
0	1	0	1	0	1	1
1	0	0	1	1	0	1
1	1	1	0	0	0	0

$$(A.B)' = A' + B'$$

● DeMorgan's Second Theorem using Truth Table

Inputs		Truth Table Outputs For Each Term				
B	A	$A+B$	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	0	1	0
1	0	1	0	1	0	0
1	1	1	0	0	0	0

$$(A+B)' = A' \cdot B'$$



THANK YOU